SAFETY DATA SHEET

Creation Date 08-Aug-2012 **Revision Date** 06-Jul-2021 **Revision Number** 9

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**Hydrazine hydrate solution 62%**
Cat No. :	**SP/2458/05**
Molecular Formula	H4 N2 . x H2 O

Unique Formula Identifier (UFI) **R7T5-KXYM-QU1P-544W**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name .
Acros Organics BVBA
Janssen Pharmaceuticalaan 3a
2440 Geel, Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road, Loughborough,
Leicestershire LE11 5RG, United Kingdom

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

Tel: 01509 231166
Chemtrec US: (800) 424-9300
Chemtrec EU: 001 (202) 483-7616

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) - **01 809 2166** (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

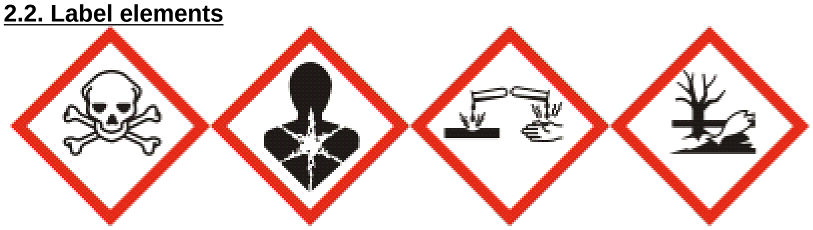
Acute oral toxicity	Category 3 (H301)
Acute dermal toxicity	Category 3 (H311)
Acute Inhalation Toxicity - Vapors	Category 2 (H330)
Skin Corrosion/Irritation	Category 1 B (H314)
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Skin Sensitization	Category 1 (H317)
Carcinogenicity	Category 1B (H350)

Environmental hazards

Acute aquatic toxicity	Category 1 (H400)
Chronic aquatic toxicity	Category 1 (H410)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word **Danger**

Hazard Statements

H301 + H311 - Toxic if swallowed or in contact with skin
H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H330 - Fatal if inhaled
H350 - May cause cancer
H410 - Very toxic to aquatic life with long lasting effects

Precautionary Statements

P201 - Obtain special instructions before use
P260 - Do not breathe dust/fume/gas/mist/vapors/spray
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician
P273 - Avoid release to the environment

2.3. Other hazards

PBT :-
This preparation contains no substance considered to be persistent, bioaccumulating nor toxic (PBT)

vPvB :-
This preparation contains no substance considered to be very persistent nor very bioaccumulating (vPvB)

Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS-No	EC-No.	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Hydrazine (hydrate)	10217-52-4	206-114-9	58 - 64	Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 2 (H330) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Skin Sens. 1 (H317) Carc. 1B (H350) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Hydrazine	302-01-2	EEC No. 206-114-9	-	Flam. Liq. 3 (H226) Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 2 (H330) Skin Corr. 1B (H314) Skin Sens. 1 (H317) Eye Dam. 1 (H318) Carc. 1B (H350) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Water	7732-18-5	231-791-2	36 - 62	-

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Hydrazine (hydrate)	Skin Corr. 1B :: C>=10% Skin Irrit. 2 :: 3%<=C<10% Eye Irrit. 2 :: 3%<=C<10%	-	-
Hydrazine	Eye Irrit. 2 (H319) :: 3%<=C<10% Skin Corr. 1B (H314) :: C>=10% Skin Irrit. 2 (H315) :: 3%<=C<10%	10	-

Components	Reach Registration Number	
Hydrazine	01-2119492624-31	(for the anhydrous form)

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

Skin Contact Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

Ingestion Call a physician immediately. Clean mouth with water.

Inhalation Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Water spray. Carbon dioxide (CO_2). Dry chemical. Chemical foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products
Nitrogen oxides (NOx), Ammonia, Hydrogen.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Avoid contact with skin, eyes and inhalation of vapors. Remove all sources of ignition. Wear protective gloves/clothing and eye/face protection. In case of inadequate ventilation wear respiratory protection.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not flush into surface water or sanitary sewer system.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Handle product only in closed system or provide appropriate exhaust ventilation. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Wear personal protective equipment/face protection.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep away from oxidizing agents. Keep away from heat, sparks and flame. Protect from light. Protect from sunlight. Store under an inert atmosphere. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) Storage Class/LGK 6.1A

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

Component	European Union	The United Kingdom	France	Belgium	Spain
Hydrazine	TWA: 0.013 mg/m³ (8h) TWA: 0.01 ppm (8h) Skin	STEL: 0.03 ppm 15 min STEL: 0.039 mg/m³ 15 min TWA: 0.01 ppm 8 hr TWA: 0.013 mg/m³ 8 hr Carc. Skin	TWA / VME: 0.1 ppm (8 heures). TWA / VME: 0.1 mg/m³ (8 heures).	TWA: 0.01 ppm 8 uren TWA: 0.013 mg/m³ 8 uren Huid	TWA / VLA-ED: 0.01 ppm (8 horas) TWA / VLA-ED: 0.013 mg/m³ (8 horas) Piel

Component	Italy	Germany	Portugal	The Netherlands	Finland
Hydrazine		Haut	TWA: 0.01 ppm 8 horas TWA: 0.013 mg/m³ 8 horas Pele	huid TWA: 0.013 mg/m³ 8 uren	TWA: 0.01 ppm 8 tunteina TWA: 0.013 mg/m³ 8 tunteina STEL: 0.05 ppm 15 minuutteina STEL: 0.07 mg/m³ 15 minuutteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Hydrazine	TRK-KZW: 0.04 ppm 15 Minuten TRK-KZW: 0.052 mg/m³ 15 Minuten Haut TRK-TMW: 0.01 ppm TRK-TMW: 0.013 mg/m³	TWA: 0.01 ppm 8 timer TWA: 0.013 mg/m³ 8 timer Hud	Haut/Peau TWA: 0.01 ppm 8 Stunden TWA: 0.013 mg/m³ 8 Stunden	STEL: 0.039 mg/m³ 15 minutach TWA: 0.013 mg/m³ 8 godzinach	TWA: 0.01 ppm 8 timer TWA: 0.01 mg/m³ 8 timer STEL: 0.03 ppm 15 minutter. value calculated STEL: 0.03 mg/m³ 15 minutter. value

					calculated Hud

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Hydrazine	TWA: 0.013 mg/m³ TWA: 0.01 ppm Skin notation	kože TWA-GVI: 0.01 ppm 8 satima. TWA-GVI: 0.013 mg/m³ 8 satima.	TWA: 0.01 ppm 8 hr. TWA: 0.013 mg/m³ 8 hr. STEL: 0.03 ppm 15 min STEL: 0.039 mg/m³ 15 min Skin	Skin-potential for cutaneous absorption TWA: 0.013 mg/m³ TWA: 0.01 ppm	TWA: 0.05 mg/m³ 8 hodinách. Potential for cutaneous absorption Ceiling: 0.025 mg/m³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Hydrazine	Nahk TWA: 0.01 ppm 8 tundides. TWA: 0.013 mg/m³ 8 tundides. STEL: 0.3 ppm 15 minutites. STEL: 0.4 mg/m³ 15 minutites.		skin - potential for cutaneous absorption TWA: 0.01 ppm TWA: 0.013 mg/m³	TWA: 0.013 mg/m³ 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 0.01 ppm 8 klukkustundum. TWA: 0.013 mg/m³ 8 klukkustundum. Skin notation Ceiling: 0.2 ppm Ceiling: 0.26 mg/m³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Hydrazine	skin - potential for cutaneous exposure TWA: 0.013 mg/m³ TWA: 0.01 ppm	TWA: 0.013 mg/m³ IPRD TWA: 0.01 ppm IPRD Oda			Skin notation TWA: 0.08 ppm 8 ore TWA: 0.1 mg/m³ 8 ore STEL: 0.8 ppm 15 minute STEL: 1 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Hydrazine	TWA: 0.1 mg/m³ 0528 Skin notation MAC: 0.3 mg/m³	TWA: 0.1 ppm 8 hodinách TWA: 0.13 mg/m³ 8 hodinách Potential for cutaneous absorption STEL: 0.5 ppm 15 minútach STEL: 0.65 mg/m³ 15 minútach	TWA: 0.01 ppm 8 urah TWA: 0.013 mg/m³ 8 urah Koža	TLV: 0.01 ppm 8 timmar. NGV TLV: 0.013 mg/m³ 8 timmar. NGV Hud	

Biological limit values
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Component	Italy	Finland	Denmark	Bulgaria	Romania
Hydrazine					Hydrazine: 200 µg/g Creatinine urine end of shift

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
See table for values

Predicted No Effect Concentration (PNEC)
See values below.

8.2. Exposure controls

Engineering Measures
Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	> 480 minutes	0.1 - 0.2 mm	Level 6	As tested under EN374-3 Determination of
Neoprene	> 480 minutes	0.38 mm	EN 374	Resistance to Permeation by Chemicals
Neoprene gloves	> 480 minutes	0.45 mm		
Viton (R)	> 480 minutes	0.30 mm		

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Organic gases and vapours filter Type K

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Half mask: EN140; Type K
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State Liquid

Appearance Colorless
Odor Ammonia-like
Odor Threshold No data available

Melting Point/Range	-65 °C / -85 °F	
Softening Point	No data available	
Boiling Point/Range	109.4 °C / 228.9 °F	
Flammability (liquid)	No data available	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	**Lower** 9.3 **Upper** 83.4	
Flash Point	73 - 91 °C / 163.4 - 195.8 °F	**Method -** Open cup
Autoignition Temperature	310 °C / 590 °F	
Decomposition Temperature	> 250°C	
pH	10.6-10.7	(1%)
Viscosity	1.26 mPa.s at 20 °C	
Water Solubility	Miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Hydrazine	-1.37	
Vapor Pressure	15 mbar @ 20 °C	
Density / Specific Gravity	1.023	
Bulk Density	Not applicable	Liquid
Vapor Density	1.1 @ 15 °C	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	H4 N2 . x H2 O
Molecular Weight	32.04
Explosive Properties	Not explosive

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

No

10.2. Chemical stability

Do not allow evaporation to dryness, Air sensitive.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	No information available.

10.4. Conditions to avoid

Exposure to air. Incompatible products.

10.5. Incompatible materials

Acids. Bases. Finely powdered metals. Halogens. nitrogen oxides (NOx). Organic materials. Peroxides. Lead. Metals. copper. Butyl rubber.

10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Ammonia. Hydrogen.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral Category 3
ATE = 244.1 mg/kg
Dermal Category 3
ATE = 588.2 mg/kg
Inhalation Category 2
ATE = 1.21 mg/l (vapours)

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Hydrazine (hydrate)	LD50 = 108 mg/kg bw (Rat) OECD TG 401	-	-
Hydrazine	LD50 = 60 mg/kg (Rat)	LD50 = 91 mg/kg (Rabbit)	570 ppm (Rat) 4 h 0.75 mg/L (Rat) 4 h
Water	-	-	-

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available
Skin Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Category 1B

Possible cancer hazard. May cause cancer based on animal data

Component	EU	UK	Germany	IARC
Hydrazine	Carc Cat. 1B		Cat. 2	Group 2A

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Test species / Duration Rat
Study result LOAEL = 0.066 mg/m^3
NOAEL = 1.92 mg/kg
Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Hydrazine	LC50: 0.28 - 1.34 mg/L, 96h static (Poecilia reticulata) LC50: 1.81 - 2.79 mg/L, 96h flow-through (Pimephales promelas) LC50: = 1.17 mg/L, 96h (Lepomis macrochirus) LC50: 0.54 - 1.31 mg/L, 96h static (Lepomis macrochirus) LC50: 0.7 - 1.3 mg/L, 96h flow-through (Lepomis macrochirus)		EC50: = 0.006 mg/L, 72h static (Pseudokirchneriella subcapitata) EC50: = 0.02 mg/L, 96h static (Pseudokirchneriella subcapitata) EC50: = 0.071 mg/L, 72h (Pseudokirchneriella subcapitata)

Component	Microtox	M-Factor
Hydrazine	EC50 = 0.01 mg/L 15 min EC50 = 0.01 mg/L 20 min EC50 = 0.02 mg/L 5 min	10

12.2. Persistence and degradability

Persistence Miscible with water, Persistence is unlikely, based on information available.

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Hydrazine	-1.37	No data available

12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment

PBT :-. This preparation contains no substance considered to be persistent, bioaccumulating nor toxic (PBT).
vPvB :-. This preparation contains no substance considered to be very persistent nor very bioaccumulating (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with high pH-value must be neutralized before discharge. Do not let this chemical enter the environment.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN2030
14.2. UN proper shipping name	HYDRAZINE, AQUEOUS SOLUTION
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	II

ADR

14.1. UN number	UN2030
14.2. UN proper shipping name	HYDRAZINE AQUEOUS SOLUTION
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	II

IATA

14.1. UN number	UN2030
14.2. UN proper shipping name	HYDRAZINE, AQUEOUS SOLUTION
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	II

14.5. Environmental hazards	Dangerous for the environment
14.6. Special precautions for user	No special precautions required
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

X = listed, Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), China (IECSC), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	IECSC	ENCS	ISHL	AICS	KECL
Hydrazine (hydrate)	-	-		-	-	-	-	X	X	X	-	-
Hydrazine	206-114-9	-		X	X	-	X	X	X	X	X	KE-19981
Water	231-791-2	-		X	X	-	X	X	X		X	KE-35400

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Hydrazine		Use restricted. See item 28. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 206-114-9 - Carcinogenic, Article 57a

https://echa.europa.eu/authorisation-list
https://echa.europa.eu/substances-restricted-under-reach
https://echa.europa.eu/candidate-list-table

Component	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Hydrazine	0.5 tonne	2 tonne

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

National Regulations

WGK Classification Water endangering class = 3 (self classification)

Component	Germany - Water Classification (VwVwS)	Germany - TA-Luft Class
Hydrazine	WGK3	

Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment
Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H301 - Toxic if swallowed
H311 - Toxic in contact with skin
H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H331 - Toxic if inhaled
H350 - May cause cancer
H410 - Very toxic to aquatic life with long lasting effects
H318 - Causes serious eye damage
H330 - Fatal if inhaled
H400 - Very toxic to aquatic life

Legend

CAS - Chemical Abstracts Service
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
PICCS - Philippines Inventory of Chemicals and Chemical Substances
ENCS - Japanese Existing and New Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
AICS - Australian Inventory of Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
TWA - Time Weighted Average
ACGIH - American Conference of Governmental Industrial Hygienists
IARC - International Agency for Research on Cancer
DNEL - Derived No Effect Level
Predicted No Effect Concentration (PNEC)
RPE - Respiratory Protective Equipment
LD50 - Lethal Dose 50%
LC50 - Lethal Concentration 50%
EC50 - Effective Concentration 50%
NOEC - No Observed Effect Concentration
POW - Partition coefficient Octanol:Water
PBT - Persistent, Bioaccumulative, Toxic
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
MARPOL - International Convention for the Prevention of Pollution from Ships
OECD - Organisation for Economic Co-operation and Development
ATE - Acute Toxicity Estimate
BCF - Bioconcentration factor
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.
Chemical incident response training.

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Revision Summary	SDS sections updated, 1, 3.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006 COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006

End of Safety Data Sheet